Stage 5

User Story: Loan Approval with EMI Calculation and Notification

As an admin, I want the system to calculate EMI after approving a loan so that the user can receive the approved loan details, EMI, total interest, and total payable amount through email.

1. Loan Application

- Create or update the Loan Application entity as per the requirements.
- Existing implementation should not be disturbed.

2. EMI Calculation Logic

Once a loan is approved, the system should calculate the EMI for each month based on:

- Loan Amount (loanAmount)
- Loan Tenure (loanDuration)
- Interest Rate (interestRate)
- Loan Type (loanType)

EMI Calculation Formula

Once a loan is approved, the system needs to calculate the EMI for each month, based on the following formula:

- **EMI Calculation Formula**:

$$EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}$$

  - `P` = Principal loan amount (`loanAmount`)
  - `R` = Monthly interest rate (Annual interest rate/12)
  - `N` = Loan tenure in months (`loanDuration`)

The system should also calculate:

- Total Interest
- Total Payable Amount

3. Loan Approval Process

Once the admin approves the loan:

1. The status of the loan should be updated to Approved.
2. Calculate the EMI based on the loan amount, interest rate, and tenure.
3. Calculate the total interest.
4. Calculate the total payable amount.
5. Store the calculated EMI in the MySQL database using JDBC.
6. Notify the user about the approved loan details via email.

4. User Email Notification

Upon loan approval, the user should receive an **actual email** containing the following information:

1. **Sanctioned Loan Amount:** The amount approved by the admin (loanAmount).
2. **Loan Type:** The type of loan applied (loanType).
3. **Loan Tenure:** The total duration of the loan (loanDuration).
4. **Applied Interest Rate:** The interest rate applied on the loan (interestRate).
5. **Monthly EMI Amount:** The EMI that the user needs to pay monthly (emiAmount).
6. **Total Interest Amount:** The total interest over the entire tenure (totalInterest).
7. **Total Payable Amount:** The sum of loan amount and interest (totalPayable).

5. Admin Dashboard for Loan Review

The admin should have options in the console-based admin menu to:

- **View All Loans:** Display all loan applications.
- **Approve/Reject Loan:** Admin can approve or reject the loan. Upon approval, the EMI should be calculated and stored.

- **Send Notification:** Upon any admin action (approval/rejection), an actual notification email should be sent to the user with the loan decision.

The application should be **console-based only**. No frontend, Spring Boot, Spring MVC, or Spring Security is required.

All database operations should be performed using **JDBC**.